Interspeech 2026
27 September - 1 October 2026, Sydney, Australia

Latency Controllable Speech Enhancement

Hiroshi Sato [1], *Takafumi Moriya* [1], *Tsubasa Ochiai* [1], *Marc Delcroix* [1]

[1] NTT, Inc. Japan
hrs.sato@ntt.com

Abstract

Streaming speech enhancement must meet latency budgets ranging from a few milliseconds (for hearing devices) to hundreds of milliseconds (for telephony/offline applications). Since latency–quality is a trade-off, an enhancer should exploit as much lookahead as the budget allows; yet most neural models are trained for a single fixed lookahead, so multiple budgets usually require multiple full models. We propose a latency-controllable speech enhancement framework consisting of a causal enhancement model augmented with Latency Control Adapters (LCAs) that allow switching the model's latency by changing how much future context it can exploit. With intra-batch multi-latency training, one LCA model supports seven latency modes; it surpasses per-latency models in enhancement quality, maintains constant inference MACs, and reduces stored parameters by 76%.

Index Terms: speech enhancement, noise reduction, latency, streaming, real-time, low latency

1. Introduction

Speech enhancement (SE) aims to estimate a clean speech signal from a noisy observation by suppressing undesired components such as environmental noise and competing sounds. SE is a core front-end technology in telecommunication, hearing assistive devices, and robust speech interfaces, where it improves perceptual quality and downstream recognition accuracy [1–5].

In many real-world deployments, SE is not judged solely by enhancement quality. The end-to-end user experience is strongly affected by the *latency* between capturing a microphone signal and rendering the enhanced output. For interactive communication (e.g., teleconferencing or VoIP), conversational quality is known to degrade as one-way delay grows to the order of a few hundred milliseconds [6–8]. For hearing assistive and acoustically transparent devices, the constraint is much stricter: when direct sound leaks and mixes with processed sound, even a few milliseconds of delay can cause audible coloration (comb-filtering) and reduced preference [9–14]. At the other end of the spectrum, one-way broadcast or offline post-processing applications may tolerate much larger latency, e.g., several hundred milliseconds or more [15, 16]. Consequently, the latency budget available to the SE module is highly application- and deployment-dependent, and may vary even within the same application due to network jitter buffers, echo cancellation pipelines, and device-side scheduling.

In general, there is a well-known trade-off between latency and enhancement performance [12, 17–19]. Non-causal models (or models with larger lookahead) can exploit future context to disambiguate speech and noise patterns, which often improves perceptual quality compared with strictly causal inference [7, 17]. When a bounded amount of future context is acceptable, increasing lookahead can yield better enhancement, and the optimal operating point depends on the latency budget. Despite this, most existing SE systems are trained and deployed under a single, pre-determined latency. Supporting multiple latency budgets in practice often requires maintaining multiple separately trained models and switching the entire model at runtime, which increases development, validation, and deployment complexity.

Motivated by these observations, we propose the concept of *latency-controllable speech enhancement*: a single SE system that can operate at multiple latency budgets and leverage available lookahead to maximize quality, while sharing the majority of parameters across budgets. The key idea is to decouple (a) a shared backbone that performs most of the enhancement computation from (b) a lightweight, latency-specific module that injects the appropriate amount of future context. As a first step toward latency-controllable SE, we investigate a mechanism for incorporating a fixed amount of future context into a streaming enhancement backbone, named the lookahead module (LAM). Based on LAM, we develop Latency Control Adapter (LCA), a lightweight adapter inserted into the SE network; by selecting an adapter corresponding to the available latency budget, the system can switch its effective lookahead at inference time without reloading the full model. Experiments show that a single LCA-equipped model can cover multiple latency budgets with improved performance, while avoiding the need to train separate models for each latency.

2. Related work

2.1. Latency performance trade-offs

Several works explicitly analyze the trade-off between latency and performance under controlled settings. Schüldt *et al.* studied real-time inference behavior of a convolutional time-domain separation/denoising network and reported that a small amount of lookahead can substantially improve denoising quality at the cost of additional delay [17]. Sato *et al.* confirmed that an additional 120 ms of lookahead substantially improved SDR and DNSMOS values in the target speaker extraction (TSE) task [19]. Schüldt *et al.* and Sato *et al.* employ an implementation that is similar to the convolution-lookahead LAM explored in our study. The key difference from our approach is that they switch the causality of the 1-d convolution layers *within* the Conv-TasNet model, whereas LAM is a lightweight DNN *appended* to the model, designed to introduce additional lookahead to an arbitrary existing system. More recently, Wu and Braun conducted a comprehensive comparison of multiple low-latency strategies under unified training and evaluation conditions, and reported a performance drop at low latencies [12].

10.21437/Interspeech.2026-2997

These studies collectively suggest that the "best" latency setting is use-case dependent, and that exploiting additional lookahead when available can improve quality.

2.2. Latency-controllable models in related speech tasks

While most SE systems are designed for a fixed latency configuration, the speech community has explored *controllable* latency and multi-mode operation in automatic speech recognition (ASR). Dual-mode ASR unifies streaming and full-context recognition by sharing weights across modes and performing joint training (optionally with in-place distillation), reducing deployment overhead and improving streaming accuracy [20]. Adaptive non-causal architectures further relax the fixed lookahead constraint by dynamically selecting when and how much future context to consume, explicitly trading off latency against accuracy during inference [21]. Accordingly, the proposed LCA in this study could also be applied to other speech tasks beyond SE, such as ASR.

3. Background

3.1. Speech enhancement with an encoder–separator–decoder backbone

We consider single-channel additive noise, $\mathbf{y} = \mathbf{x} + \mathbf{n} \in \mathbb{R}^T$, where $\mathbf{y}$, $\mathbf{x}$, and $\mathbf{n}$ are the observed, clean, and noise signals, respectively. T represents the number of samples. A DNN-based SE model $\texttt{SE}(\cdot)$ estimates $\hat{\mathbf{x}}$ from $\mathbf{y}$ as $\hat{\mathbf{x}} = \texttt{SE}(\mathbf{y}; \boldsymbol{\theta})$, where $\boldsymbol{\theta}$ denotes the learnable parameters.

Many modern SE systems adopt an encoder–separator–decoder structure [22–26]:

$$\mathbf{Z}_{\text{enc}} = \texttt{Encoder}(\mathbf{y}; \boldsymbol{\theta}_{\text{enc}}) \in \mathbb{R}^{T' \times D}, \tag{1}$$

$$\mathbf{Z}_{\text{sep}} = \texttt{Separator}(\mathbf{Z}_{\text{enc}}; \boldsymbol{\theta}_{\text{sep}}) \in \mathbb{R}^{T' \times D}, \tag{2}$$

$$\hat{\mathbf{x}} = \texttt{Decoder}(\mathbf{Z}_{\text{sep}}; \boldsymbol{\theta}_{\text{dec}}), \tag{3}$$

where T' and D denote the number of frames (sequence length) and feature dimension in the latent representation, respectively. The latent representations at the output of the encoder and separator are denoted by $\mathbf{Z}_{\text{enc}}$ and $\mathbf{Z}_{\text{sep}}$, respectively. In time–frequency domain SE, `Encoder/Decoder` are typically STFT/iSTFT and the separator estimates a mask or mapping in the STFT domain [27]. In time-domain SE, `Encoder/Decoder` are commonly learned 1-D convolutional filterbanks [28]. All parameters $\boldsymbol{\theta} = \{\boldsymbol{\theta}_{\text{enc}}, \boldsymbol{\theta}_{\text{sep}}, \boldsymbol{\theta}_{\text{dec}}\}$ are trained jointly by minimizing a criterion between $\hat{\mathbf{x}}$ and $\mathbf{x}$.

3.2. Algorithmic latency in neural SE

We focus on *algorithmic latency*, i.e., the delay induced by buffering and the use of future context (excluding compute time). Let L_{len} and L_{shift} be the analysis window length and hop size (in samples), and f_s be the sampling rate. For time-domain convolutional encoders, L_{len} and L_{shift} correspond to kernel size and stride. The encoder must accumulate L_{len} samples before producing the first latent frame, yielding latency in seconds τ_{enc} as $\tau_{\text{enc}} = L_{\text{len}}/f_s$. Note that some studies report latency without explicitly accounting for this buffering term, so care is needed when comparing them [29].

In general, the separator may exploit future latent frames. Let $T_{\text{la,sep}}$ be the number of future frames used by the separator (in the latent frame rate). This introduces additional latency in seconds τ_{sep} as $\tau_{\text{sep}} = T_{\text{la,sep}} \cdot L_{\text{shift}}/f_s$. We call a system *causal* if the separator itself uses no future frames ($T_{\text{la,sep}} = 0$), even though τ_{enc} is still present. The total algorithmic latency τ is expressed as $\tau = \tau_{\text{enc}} + \tau_{\text{sep}}$.

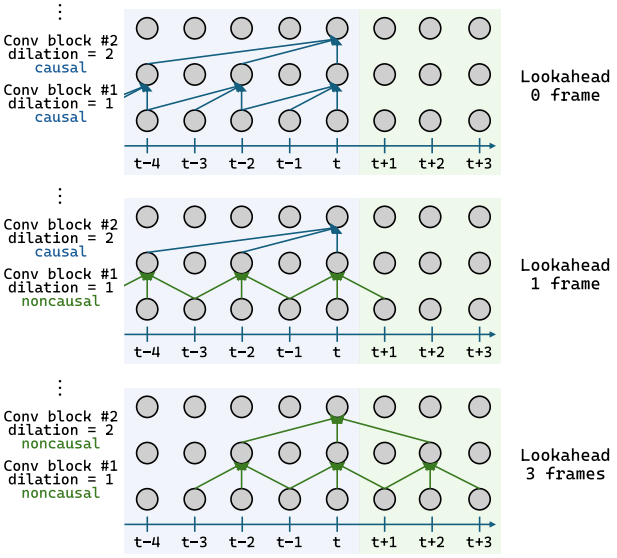


Figure 1: *Illustration of lookahead realized by convolution lookahead blocks (kernel size 3). Switching a block from causal to non-causal adds dilation-dependent right context.*

3.3. Lookahead module (LAM)

We insert a *lookahead module* (LAM) into a causal backbone (where the separator uses no future frames) or semi-causal backbone (with a limited amount of latency) to give it access to a specified additional amount of lookahead. For exposition, assume the separator is composed of R blocks and denote the intermediate representation after the i-th block as $\mathbf{Z}_i \in \mathbb{R}^{T' \times D}$ with $\mathbf{Z}_0 = \mathbf{Z}_{\text{enc}}$ and $\mathbf{Z}_R = \mathbf{Z}_{\text{sep}}$. If LAM is inserted after r-th block, LAM transforms $\mathbf{Z}_r$ into $\mathbf{Z}'_r$ by consuming past and future frames:

$$\mathbf{Z}'_r[t] = \texttt{LAM}\left(\mathbf{Z}_r[t - T_{\text{lb}} : t + T_{\text{la}}]; \boldsymbol{\theta}_{\text{lam}}\right), \tag{4}$$

where T_{lb} and T_{la} are the lookback/lookahead (in frames). Therefore, LAM introduces additional algorithmic latency τ_{lam} as $\tau_{\text{lam}} = T_{\text{la}} \cdot L_{\text{shift}}/f_s$. LAM parameters $\boldsymbol{\theta}_{\text{lam}}$ are trained jointly with the backbone parameters $\boldsymbol{\theta}$.

One simple LAM instantiation is a stack of dilated 1-D convolutional blocks. We use kernel size 3 and X blocks whose dilations are $d = 2^0, 2^1, \ldots, 2^{X-1}$. Each block can operate either (i) *causally* (no right context) or (ii) *non-causally* (symmetric context), which enables controlling the total lookahead. Let $\mathbf{b} = \{b_i\}_{i=1}^X$ and $b_i \in \{0, 1\}$ indicate whether the i-th dilated block is non-causal. With kernel size 3, a non-causal block with dilation 2^i introduces 2^i frames of right context, and the total lookahead becomes:

$$T_{\text{la}} = \sum_{i=1}^{X} b_i \, 2^{i-1}. \tag{5}$$

Figure 1 illustrates how different binary patterns yield different lookahead. In addition, each 1-D convolution layer is implemented with depthwise-separable convolution (as in Conv-TasNet) [28], for improved efficiency.

In the literature, a single 1-D convolutional layer was used to introduce lookahead [30]. By extending this method to dilated 1-D convolutions, we can achieve reduced computational

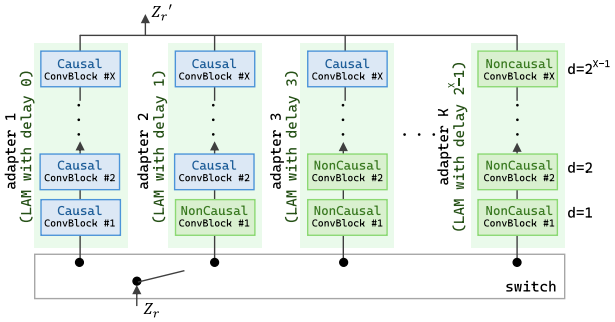


Figure 2: *Conceptual diagram of the proposed Latency Control Adapter (LCA). Each adapter corresponds to a different convolution-lookahead configuration, and a switch selects the adapter at inference time.*

cost, particularly when a long-term future context is desired. [1]

4. Proposed method

4.1. Latency Control Adapter (LCA)

We aim to realize *latency-controllable SE*, where a single model supports multiple lookahead budgets and switches its latency at runtime without reloading the full network. To this end, we propose a *Latency Control Adapter (LCA)*: a bank of relatively lightweight adapters inserted into the separator, where each adapter is an LAM configured for a specific lookahead. Figure 2 shows the conceptual diagram. At inference time, we select an adapter according to the available latency budget.

4.1.1. Parameter sharing across adapters

A simple implementation learns a separate parameter set for each adapter, resulting in a linear increase in the number of parameters with the number of latency modes. Formally, let the number of adapters and model parameters per adapter be K and M, respectively. Then, the total number of model parameters in LCA is $K \cdot M$, which raises scalability concerns when K is large. To address this, we consider further reducing the total number of parameters in LCA by sharing model parameters across adapters. In this study, we evaluate a scheme where, for the i-th 1-D Conv Block, parameters are shared across adapters for causal blocks and, separately, shared across adapters for non-causal blocks. In this case, the total number of model parameters of LCA becomes $2 \cdot M$, which has the advantage of being independent of the number of adapters K.

4.1.2. Intra-batch multi-latency training

We promote knowledge sharing across latency modes by *intra-batch multi-latency training*. For each mini-batch, we run the same input through all K adapters and average the losses. Using the SNR objective, we minimize

$$\mathcal{L} = \frac{1}{K} \sum_{i=1}^{K} \left(-\mathtt{SNR}(\hat{\mathbf{x}}^{(i)}, \mathbf{x}) \right), \tag{6}$$

where $\hat{\mathbf{x}}^{(i)}$ is the enhanced waveform produced by selecting adapter i. All parameters, including the shared backbone and adapter parameters, are optimized jointly.

[1] In preliminary experiments, we confirmed that, for achieving an equivalent lookahead, using dilated 1-D convolutions yields better enhancement performance than using a single convolution layer.

Table 1: *Computational cost to support seven latency modes.*

Systems	MACs [G/s]	Total #params [M]
(A-1) Causal	1.31	13.3
(B-1) Convolution lookahead (7 models)	1.46	104.1
(C-1) LCA (shared adapter params)	1.46	16.5
(C-2) LCA (no sharing)	1.46	24.5

5. Experiments

5.1. Experimental setup

5.1.1. Data

Training and evaluation were conducted on simulated noisy mixtures generated from clean speech in LibriSpeech [31] and noise clips from the DNS4 challenge dataset [32]. All audio was sampled at $f_s = 16$ kHz. Noisy mixtures were created by adding a randomly selected noise segment at an SNR uniformly sampled from $[0, 20]$ dB. We generated 50,000, 3,000, and 2,000 mixtures for training, development, and evaluation, respectively.

5.1.2. System configuration

As the SE backbone, we used a time-domain Conv-TasNet [28], which offers a strong accuracy–efficiency trade-off, is real-time capable on CPU, and remains a well-established baseline with many extensions [19,33–36]. Following the notation in [28], we set B $= 256$, R $= 4$, X $= 8$, H $= 512$, and P $= 3$. We evaluated two encoder–decoder configurations to cover both moderate and ultra-low base latency regimes: (i) a *standard latency* setting with encoder kernel/stride $(L_{\text{len}}, L_{\text{shift}}) = (320, 160)$ and 4096 encoder filters, yielding $\tau_{\text{enc}} = 20$ ms; (ii) an *ultra-low latency* setting with $(L_{\text{len}}, L_{\text{shift}}) = (80, 40)$ and 1024 encoder filters, yielding $\tau_{\text{enc}} = 5$ ms. Unless otherwise stated, results use the standard latency setting.

For both LAM and LCA, we considered $K = 7$ lookahead budgets, $T_{\text{la}} \in \{0, 1, 3, 7, 15, 31, 63\}$ (in frames) In the standard latency setting, this yields total latencies $\tau \in \{20, 30, 50, 90, 170, 330, 650\}$ ms; in the ultra-low latency setting, $\tau \in \{5, 7.5, 12.5, 22.5, 42.5, 82.5, 162.5\}$ ms. We used a stack of X=6 dilated convolution blocks (kernel size 3; dilations 2^0–2^5). To realize the 7 lookahead candidates, we used the monotone binary patterns of causality $\mathbf{b}$ (defined in Section 3.3), $\mathbf{b} \in \{\mathbf{0}, [1, 0, \ldots], [1, 1, 0, \ldots], \ldots, \mathbf{1}\}$. The insertion position of LAM and LCA is after the $r = 4$-th block of Conv-TasNet within $R = 4$ blocks, i.e., after the separator. [2]

5.1.3. Training and evaluation

All systems were trained under the same recipe, using the negative SNR objective. For LCA, we applied intra-batch multi-latency training, i.e., averaging losses over all K adapters in each mini-batch. At inference time, only one adapter is selected according to the target latency budget. As objective metrics, we report SDR [37] and DNSMOS P.835 OVRL [38]. To characterize efficiency, we report (i) multiply–accumulate operations per second of audio (MACs, in G/s), and (ii) the total number of learnable parameters required to support all seven latency modes (Table 1).

[2] We preliminarily tested three insertion positions within the separator: (i) before the separator, (ii) in the middle of the separator (after the second repeat), and (iii) after the separator. Condition (iii) yielded the best results.

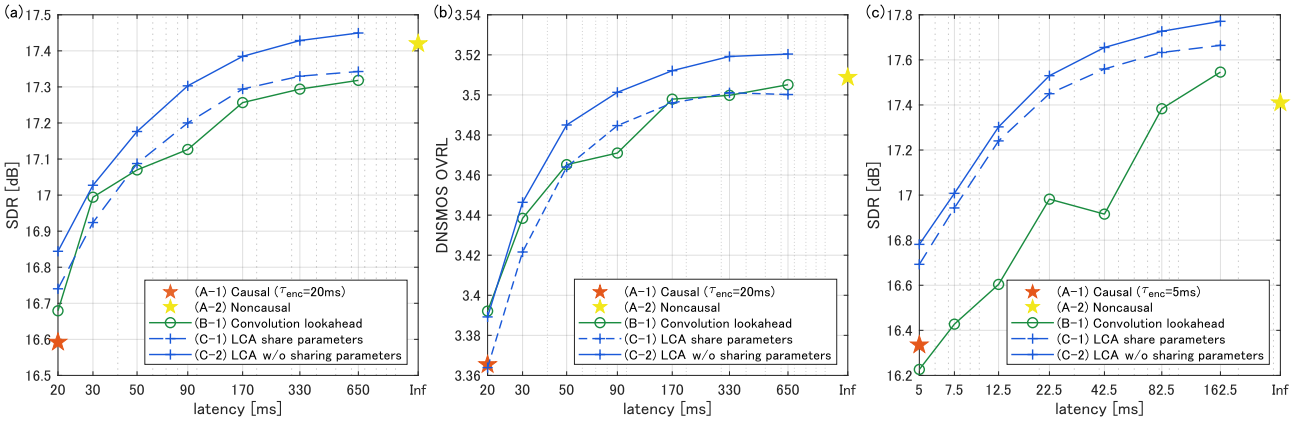


Figure 3: *Latency–performance trade-offs in (a) and (b) the standard encoder setting ($\tau_{\mathrm{enc}} = 20$ ms) and in (c) the low-latency encoder setting ($\tau_{\mathrm{enc}} = 5$ ms). (a) and (c) show SDR and (b) shows DNSMOS P.835 OVRL.*

Table 2: *Ablation study on LCA training and design*

Systems	Add. lookahead [ms]		
	10	70	310
(C-2) LCA (no sharing)	**17.03**	**17.30**	**17.43**
w/o joint training of the whole model	16.67	16.73	16.77
w/o causal blocks in LCA	16.75	17.00	17.11
w/o intra-batch multi-latency training	16.80	17.08	17.22

5.2. Experimental results

Figure 3 (a) and (b) compares latency–performance trade-offs between baseline systems and LCA. We consider: (A-1) a causal Conv-TasNet, (A-2) a non-causal Conv-TasNet, (B-1) *multi-model* convolution lookahead with LAM, where a separate model is trained for each latency, and (C-1)/(C-2) *single-model* LCA, where latency is switched by selecting an adapter at inference time. (C-1) shares adapter parameters across latency modes, while (C-2) uses separate parameters per adapter. All streaming systems exhibit a clear latency–performance trade-off: larger lookahead yields higher SDR and DNSMOS. Notably, (C-2) achieves higher SDR and DNSMOS than (B-1) across most latency points, despite using a single shared backbone; the average SDR gain over the seven operating points is 0.13 dB in this setting. We attribute this improvement to intra-batch multi-latency training, which encourages knowledge distillation across latency modes during joint optimization.

Table 1 summarizes compute and parameter costs. Adding a LAM increases MACs compared with the causal baseline. This runtime cost is identical for (B-1) and the LCA variant because only one lookahead path (either a LAM or a selected adapter) is executed at inference time. In contrast, the parameter cost differs substantially when supporting multiple latency modes. To cover seven budgets, (B-1) requires storing seven separately trained full models (104.1 M parameters total). LCA shares the backbone across modes and stores only the adapter bank, reducing total parameters to 16.5 M (C-1) or 24.5 M (C-2). The shared-parameter design (C-1) keeps the adapter parameter overhead independent of the number of supported latency modes, while (C-2) provides better quality at the expense of more parameters.

Figure 3 (c) repeats the SDR analysis under the ultra-low latency encoder configuration ($\tau_{\mathrm{enc}} = 5$ ms). The same qualitative trend holds, and LCA remains effective in this stricter regime. When compared at the same amount of latency, the ultra-low latency condition yields better performance than the standard latency condition because the finer analysis window shift in the Conv-TasNet encoder/decoder enables a more precise projection of the waveform into the feature domain at the expense of additional MACs [19, 28, 39]. Compared with the multi-model baseline (B-1), (C-2) yields a larger improvement in SDR (0.53 dB on average across the seven operating points), suggesting that joint multi-latency training is particularly beneficial when the base latency is very small. It is worth noting that under the 5 ms latency setting, system (C-2) improves SDR by 0.45 dB over the causal model (A-1). This gain is attributable to knowledge distillation from a larger lookahead system; consequently, even in scenarios where only a single low-latency model is required, the proposed LCA can still deliver improved enhancement performance.

Table 2 investigates factors contributing to LCA effectiveness (using (C-2) as the reference). First, training only the LCA on top of a pretrained backbone while freezing the rest of the model degrades performance substantially, indicating that adapting shared backbone parameters is important when accommodating multiple latency modes. Second, removing the causal blocks inside LCA (i.e., constructing adapters only from the non-causal blocks required for each lookahead) also reduces performance, suggesting that keeping a consistent adapter structure across modes stabilizes optimization. Finally, replacing intra-batch multi-latency training with a weaker strategy (randomly selecting a single adapter per mini-batch) leads to lower SDR, supporting the role of joint multi-latency training for cross-latency knowledge sharing.

6. Conclusion

This paper introduces the concept of *latency-controllable speech enhancement*, which adapts the lookahead to meet latency requirements and leverages the available future context for more effective inference. To realize this, we proposed a *Latency Control Adapter (LCA)* based on a *lookahead module (LAM)* using dilated convolutions, and enabled runtime latency switching by selecting a latency-specific adapter. Compared with a baseline that prepares separate models for each latency, the proposed LCA shares the backbone parameters across latency budgets and switches only the lightweight adapter, reducing the total number of model parameters while improving enhancement performance at each latency.

7. Generative AI use disclosure

This manuscript was drafted by the authors and edited and polished with the assistance of a generative AI tool (ChatGPT). All experimental design, implementation, and analysis were conducted by the authors, who assume full responsibility for the content.

8. References

[1] C. K. A. Reddy, V. Gopal, R. Cutler, E. Beyrami, R. Cheng, H. Dubey, S. Matusevych, R. Aichner, A. Aazami, S. Braun, P. Rana, S. Srinivasan, and J. Gehrke, "The INTERSPEECH 2020 deep noise suppression challenge: Datasets, subjective testing framework, and challenge results," in *Proc. INTERSPEECH*, 2020.

[2] Microsoft DNS Challenge Organizers, "Fifth deep noise suppression challenge (ICASSP 2023)," Project website, 2023, accessed: 2026-02-26. [Online]. Available: https://www.microsoft.com/en-us/research/academic-program/deep-noise-suppression-challenge-icassp-2023/

[3] S. T. Yousif and B. M. Mahmmod, "Speech enhancement algorithms: A systematic literature review," *Algorithms*, vol. 18, no. 5, p. 272, 2025.

[4] H. Sato, R. Masumura, T. Ochiai, M. Delcroix, T. Moriya, T. Ashihara, K. Shinayama, S. Mizuno, M. Ihori, T. Tanaka, and N. Hojo, "Downstream Task Agnostic Speech Enhancement with Self-Supervised Representation Loss," in *Proc. Annual Conference of the International Speech Communication Association*, 2023, pp. 854–858.

[5] H. Sato, T. Ochiai, M. Delcroix, T. Moriya, T. Ashihara, and R. Masumura, "Generic speech enhancement with self-supervised representation space loss," *Frontiers in Signal Processing*, vol. 5, p. 1587969, 2025.

[6] "Recommendation ITU-T G.114: One-way transmission time," International Telecommunication Union, ITU-T Recommendation, 2003. [Online]. Available: https://www.itu.int/rec/T-REC-G.114

[7] S. Drgas, "A survey on low-latency DNN-based speech enhancement," *Sensors*, vol. 23, no. 3, p. 1380, 2023.

[8] H. Dubey, V. Gopal, R. Cutler, A. Aazami, S. Braun *et al.*, "The ICASSP 2023 deep noise suppression challenge," *IEEE Open Journal of Signal Processing*, vol. 5, pp. 725–737, 2024.

[9] M. A. Akeroyd, W. Bailey, J. Barker, T. J. Cox, J. F. Culling, S. Graetzer, G. Naylor, Z. Podwińska, and Z. Tu, "The 2nd clarity enhancement challenge for hearing aid speech intelligibility enhancement: Overview and outcomes," in *International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.

[10] J. Agnew and J. M. Thornton, "Just noticeable and objectionable group delays in digital hearing aids," *Journal of the American Academy of Audiology*, vol. 11, no. 06, pp. 330–336, 2000.

[11] D. Lelic, G. Stiefenhofer, E. Lundorff, and T. Neher, "Hearing aid delay in open-fit devices: Preferred sound quality in normal-hearing listeners and hearing aid users," *JASA Express Letters*, vol. 2, no. 10, p. 104803, 2022.

[12] H. Wu and S. Braun, "Ultra-low latency speech enhancement – a comprehensive study," in *Proc. ICASSP*, 2025.

[13] Z.-Q. Wang, G. Wichern, S. Watanabe, and J. Le Roux, "Stft-domain neural speech enhancement with very low algorithmic latency," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 397–410, 2022.

[14] S. J. Yang, S. Wisdom, C. Gnegy, R. F. Lyon, and S. Savla, "Listening with googlears: Low-latency neural multiframe beamforming and equalization for hearing aids." in *Proc. Interspeech*, 2022, pp. 3939–3943.

[15] A. Bentaleb, M. Lim, M. N. Akcay, A. C. Begen, S. Hammoudi, and R. Zimmermann, "Toward one-second latency: Evolution of live media streaming," *IEEE Communications Surveys & Tutorials*, 2025.

[16] W. Lasecki, C. Miller, A. Sadilek, A. Abumoussa, D. Borrello, R. Kushalnagar, and J. Bigham, "Real-time captioning by groups of non-experts," in *Proceedings of the 25th annual ACM symposium on User interface software and technology*, 2012, pp. 23–34.

[17] C. Schüldt, S. Sonning, H. Erdogan, and S. Wisdom, "Performance study of a convolutional time-domain audio separation network for real-time speech denoising," in *Proc. ICASSP*, 2020, pp. 831–835.

[18] B. Patton, J. Skoglund, J. Thorpe, J. Hershey, K. Wilson, M. Chinen, R. F. Lyon, and R. A. Saurous, "Exploring tradeoffs in models for low-latency speech enhancement," in *Proceedings of the 16th International Workshop on Acoustic Signal Enhancement*, 2018.

[19] H. Sato, T. Moriya, M. Mimura, S. Horiguchi, T. Ochiai, T. Ashihara, A. Ando, K. Shinayama, and M. Delcroix, "SpeakerBeam-SS: Real-time Target Speaker Extraction with Lightweight Conv-TasNet and State Space Modeling," in *Proc. Annual Conference of the International Speech Communication Association*, 2024, pp. 5033–5037.

[20] J. Yu, W. Han, A. Gulati, C.-C. Chiu, B. Li, T. N. Sainath, Y. Wu, and R. Pang, "Dual-mode ASR: Unify and improve streaming ASR with full-context modeling," in *International Conference on Learning Representations (ICLR)*, 2021. [Online]. Available: https://openreview.net/forum?id=Pz_dcqfcKW8

[21] G. P. Strimel, Y. Xie, B. King, M. Radfar, A. Rastrow, and A. Mouchtaris, "Lookahead when it matters: Adaptive non-causal transformers for streaming neural transducers," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 202, 2023.

[22] Y. Luo and N. Mesgarani, "TasNet: time-domain audio separation network for real-time, single-channel speech separation," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing*, 2018, pp. 696–700.

[23] E. Kim and H. Seo, "SE-Conformer: Time-Domain Speech Enhancement Using Conformer," in *Proc. Annual Conference of the International Speech Communication Association*, 2021, pp. 2736–2740.

[24] Y. Hu, Y. Liu, S. Lv, M. Xing, S. Zhang, Y. Fu, J. Wu, B. Zhang, and L. Xie, "DCCRN: Deep Complex Convolution Recurrent Network for Phase-Aware Speech Enhancement," in *Proc. Annual Conference of the International Speech Communication Association*, 2020, pp. 2472–2476.

[25] H. Schroter, A. N. Escalante-B, T. Rosenkranz, and A. Maier, "Deepfilternet: A low complexity speech enhancement framework for full-band audio based on deep filtering," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 7407–7411.

[26] E. Tzinis, Z. Wang, and P. Smaragdis, "Sudo rm-rf: Efficient networks for universal audio source separation," in *2020 IEEE 30th International Workshop on Machine Learning for Signal Processing (MLSP)*. IEEE, 2020, pp. 1–6.

[27] Y. Xu, J. Du, L.-R. Dai, and C.-H. Lee, "A regression approach to speech enhancement based on deep neural networks," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 23, no. 1, pp. 7–19, 2015.

[28] Y. Luo and N. Mesgarani, "Conv-TasNet: Surpassing ideal time–frequency magnitude masking for speech separation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 27, no. 8, pp. 1256–1266, 2019.

[29] H. Wu and S. Braun, "Ultra-low latency speech enhancement – a comprehensive study," *arXiv preprint arXiv:2409.10358*, 2024.

[30] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, "Mobilenets: Efficient convolutional neural networks for mobile vision applications," *arXiv preprint arXiv:1704.04861*, 2017.

[31] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "LibriSpeech: An ASR corpus based on public domain audio books," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing*, 2015, pp. 5206–5210.

[32] C. K. Reddy, H. Dubey, K. Koishida, A. Nair, V. Gopal, R. Cutler, S. Braun, H. Gamper, R. Aichner, and S. Srinivasan, "INTERSPEECH 2021 deep noise suppression challenge," in *Proc. Annual Conference of the International Speech Communication Association*, 2021, pp. 2796–2800.

[33] W. Zhang, J. Shi, C. Li, S. Watanabe, and Y. Qian, "Closing the gap between time-domain multi-channel speech enhancement on real and simulation conditions," in *2021 IEEE workshop on applications of signal processing to audio and acoustics (WASPAA)*. IEEE, 2021, pp. 146–150.

[34] D. Lee, S. Kim, and J.-W. Choi, "Inter-channel conv-tasnet for multichannel speech enhancement," *arXiv preprint arXiv:2111.04312*, 2021.

[35] C. Li, J. Shi, W. Zhang, A. S. Subramanian, X. Chang, N. Kamo, M. Hira, T. Hayashi, C. Boeddeker, Z. Chen *et al.*, "Espnet-se: End-to-end speech enhancement and separation toolkit designed for asr integration," in *2021 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2021, pp. 785–792.

[36] M. Pariente, S. Cornell, J. Cosentino, S. Sivasankaran, E. Tzinis, J. Heitkaemper, M. Olvera, F.-R. Stöter, M. Hu, J. M. Martín-Doñas, D. Ditter, A. Frank, A. Deleforge, and E. Vincent, "Asteroid: The PyTorch-Based Audio Source Separation Toolkit for Researchers," in *Proc. Interspeech*, 2020, pp. 2637–2641.

[37] E. Vincent, R. Gribonval, and C. Févotte, "Performance measurement in blind audio source separation," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 14, no. 4, pp. 1462–1469, 2006.

[38] C. K. Reddy, V. Gopal, and R. Cutler, "Dnsmos p. 835: A non-intrusive perceptual objective speech quality metric to evaluate noise suppressors," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing*, 2022, pp. 886–890.

[39] M. Thakker, S. E. Eskimez, T. Yoshioka, and H. Wang, "Fast Real-time Personalized Speech Enhancement: End-to-End Enhancement Network (E3Net) and Knowledge Distillation," in *Proc. Annual Conference of the International Speech Communication Association*, 2022, pp. 991–995.